

Can a vision model identify every item in a fashion screenshot?

A benchmark of item-level extraction: detect each product, classify it, describe it, and write the query to find it.



WOMEN'S ETHNIC



FUSION



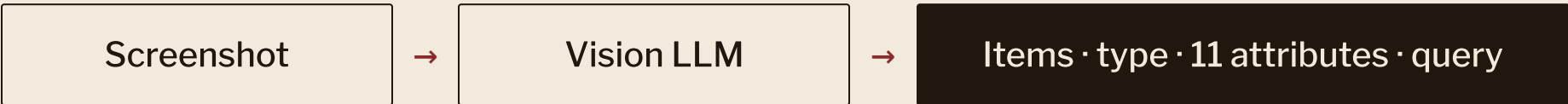
GRAPHIC TEE

OBJECTIVE	Identify every purchasable item: type, attributes, search query
TEST SET	100 screenshots · 8 categories · 46 celebrity, 54 everyday
MODELS	11 vision LLMs · Google, OpenAI, Anthropic, Alibaba
ACCESS	One API through OpenRouter
FRAMEWORK	FNDS: Finds · Names · Describes · Search, scored 0–100

One image in. Structured JSON out.

PROCESS Single-pass extraction

PIPELINE



One call per image. Latency = median end-to-end time per call, n = 100.

GROUND TRUTH

Two frontier models label each image independently: Gemini 3.1 Pro and Claude Sonnet 5.5. Only their consensus is scored, covering 92% of items.

Audit by GPT-5.6 Terra on 20 images: 100% of items valid, 97.8% of attributes correct.

The FNDS Framework

$0.25F + 0.25N + 0.30D + 0.20S$

F	Finds	25%	Item recall against ground truth
N	Names	25%	Correct product type: kurta ≠ kurti
D	Describes	30%	Attribute accuracy: colour, pattern, fabric, fit...
S	Search	20%	Query contains the key search terms

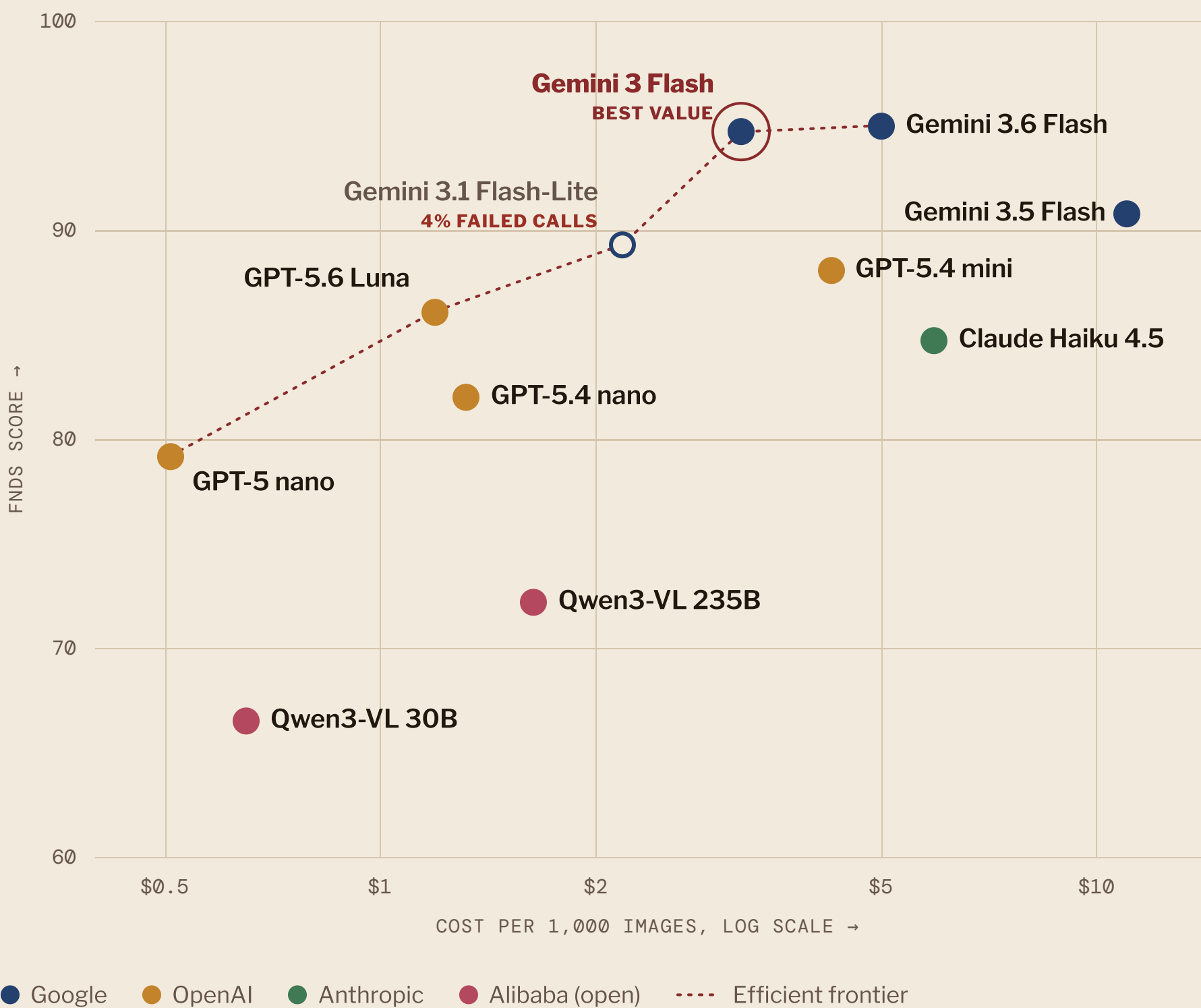
Near misses earn 0.5: trousers read as leggings, navy read as black.

Gemini Flash models lead on FNDS.

MODEL	FNDS	F	N	D	S	\$ / 1K	LATENCY
Gemini 3.6 Flash Google	95.0	94.8	98.2	94.3	92.1	\$5.01	6.8s
Gemini 3 Flash Best value	94.7	98.8	98.4	93.5	91.3	\$3.19	4.7s
Gemini 3.5 Flash Google	90.8	88.1	95.3	92.8	86.5	\$11.05	5.0s
Gemini 3.1 Flash-Lite 4% failed calls	89.3	93.1	98.1	91.9	88.4	\$2.18	4.5s
GPT-5.4 mini OpenAI	88.1	93.9	95.0	82.7	80.2	\$4.26	5.9s
GPT-5.6 Luna OpenAI	86.1	97.3	95.5	74.8	77.3	\$1.19	8.0s
Claude Haiku 4.5 Anthropic	84.7	90.5	94.5	80.2	72.2	\$5.94	5.6s
GPT-5.4 nano OpenAI	82.0	94.4	92.2	71.4	73.9	\$1.32	6.2s
GPT-5 nano OpenAI	79.2	84.3	89.5	73.0	74.1	\$0.51	10.7s
Qwen3-VL 235B Alibaba (open)	72.2	71.4	76.1	73.2	70.8	\$1.64	22.2s
Qwen3-VL 30B Alibaba (open)	66.5	70.1	70.5	66.0	61.5	\$0.65	20.7s

F Finds · N Names · D Describes · S Search. FNDS = 0.25F + 0.25N + 0.30D + 0.20S. Latency: median per image via OpenRouter. Gaps under 5 points are within noise at n = 100.

The most expensive model isn't the best one.



SWEET SPOT

Gemini 3 Flash: 94.7 at \$3.19, top-left of the map.

OVERPAYING

Gemini 3.5 Flash: 3.5× the cost, 4 points lower.

FLOOR

Qwen3-VL sits cheapest-and-lowest; no open model reaches 75.

Quality saturates near \$3 per 1,000 images.

MODEL	\$ PER 1K IMAGES	F	N	D	S	FNDS
GPT-5 nano	\$0.51	84	90	73	74	79
Qwen3-VL 30B	\$0.65	70	70	66	62	66
GPT-5.6 Luna	\$1.19	97	96	75	77	86
GPT-5.4 nano	\$1.32	94	92	71	74	82
Qwen3-VL 235B	\$1.64	71	76	73	71	72
Gemini 3.1 Flash-Lite	\$2.18	93	98	92	88	89
Gemini 3 Flash	\$3.19	99	98	94	91	95
GPT-5.4 mini	\$4.26	94	95	83	80	88
Gemini 3.6 Flash	\$5.01	95	98	94	92	95
Claude Haiku 4.5	\$5.94	90	94	80	72	85
Gemini 3.5 Flash	\$11.05	88	95	93	86	91

F + N

Near-solved. Most models score 90+.

D + S

Where models separate.

PRICE

3.5× the cost of Gemini 3 Flash buys 4 points less.

Sorted by cost. Bar on log scale. Darker = higher score.

Gemini 3 Flash.

Works best on cost and median latency.

94.7

FNDS

\$3.19

per 1,000 images

4.7s

median latency

Ties the top score at 64% of its cost, and beats Gemini 3.5 Flash by 4 points at 29% of the cost. Highest item recall (98.8) and type accuracy (98.4).

The trade-off

	Gemini 3 Flash	GPT 5.4 mini · 5 nano	GEMINI vs GPT vs QWEN Qwen3-VL 235B · 30B, open
FNDS	94.7	79-88	67-72
\$ / 1K	\$3.19	\$0.51-4.26	\$0.65-1.64
LATENCY	4.7s	5.9-10.7s	~20s
BEST FOR	Live extraction at scale	Lowest API cost	Offline tagging and self-hosting

Via API, GPT-5 nano beats Qwen on both price (\$0.51) and score (79). Qwen earns its place only when images must stay in-house.